

Implementing Initial Security Policies

Hands-on Lab
Week 2, Day 1
Collaborators:

Part 1: Map Policy to Controls

1. Before changing anything, create a table mapping each of the six policy lines above to the specific Linux mechanism you'll use to enforce it.

Policy line	Linux Mechanism	Configuration file/command
Passwords must be at least 12 characters and enforce complexity requirements	pam_pwquality	sudo apt install libpam-pwquality -y sudo cp /etc/security/pwquality.conf /etc/security/pwquality.conf.bak sudo nano /etc/security/pwquality.conf
Accounts must lock after 5 consecutive failed login attempts, for a minimum of 15 minutes	pam_faillock	Deny = 5 Unlock_time = 900
All authentication attempts (success and failure) must be logged and retained	auditd	/etc/audit/auditd.conf /var/log/audit/audit.log
Users must see an authorized-use warning banner before authenticating	SSH Banner	/etc/ issue.net /etc/ssh/sshhd_config
Every active account must map to a current, authorized user - no orphaned or default accounts	Manual account review	cat /etc/passwd
Critical system files must be monitored for unauthorized changes	AIDE	/var/lib/aide/aide.db

2. In table
3. Review the current account list (cat /etc/passwd) and flag any accounts that appear to violate the “no orphaned accounts” requirement.

Two accounts should be reviewed for the “no orphaned/default accounts” requirement:

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- oldservice – Uses /bin/sh and has its own home directory at /home/oldservice. The account appears to be an older service account and should be verified to confirm it is still required.
- logreview – Uses /bin/bash and has its own home directory at /home/logreview. This account should be verified to ensure it still has a valid owner and business purpose.

```
student@LinuxTargetVM2:~$ pwd
/home/student
student@LinuxTargetVM2:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uidd:x:103:103::/run/uidd:/usr/sbin/nologin
tss:x:104:104:TPM software stack,,,:/var/lib/tpm:/bin/false
sshd:x:105:65534::/run/sshd:/usr/sbin/nologin
pollinate:x:106:1::/var/cache/pollinate:/bin/false
tcpdump:x:107:108::/nonexistent:/usr/sbin/nologin
landscape:x:108:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:989:989:User for polkitd:/:/usr/sbin/nologin
_chrony:x:109:113:Chrony daemon,,,:/var/lib/chrony:/usr/sbin/nologin
student:x:1000:1000:Ubuntu:/home/student:/bin/bash
oldservice:x:1001:1001::/home/oldservice:/bin/sh
logreview:x:1002:1002:logreview,,,:/home/logreview:/bin/bash
postfix:x:110:115::/var/spool/postfix:/usr/sbin/nologin
_aide:x:988:988:Advanced Intrusion Detection Environment:/var/lib/aide:/usr/sbin/nologin
```

Part 2: Password and Account Lockout Policy

1. Install and configure `pam_pwquality` to require a minimum password length of 12 characters with complexity enforced.

```
student@LinuxTargetVM2:~$ pwd
/home/student
student@LinuxTargetVM2:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
10 packages can be upgraded. Run 'apt list --upgradable' to see them.
student@LinuxTargetVM2:~$ sudo apt install libpam-pwquality
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libpam-pwquality is already the newest version (1.4.5-3build1).
The following packages were automatically installed and are no longer required:
  libmbim-utils libqmi-utils libtcl8.6 tcl tcl8.6 usb-modeswitch usb-modeswitch-data
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 10 not upgraded.
```

```
student@LinuxTargetVM2:~$ sudo cp /etc/security/pwquality.conf /etc/security/pwquality.conf.bak
```

Before:

```
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
# minlen = 8
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
# dcredit = 0
#
# The maximum credit for having uppercase characters in the new password.
# If less than 0 it is the minimum number of uppercase characters in the new
# password.
# ucredit = 0
#
# The maximum credit for having lowercase characters in the new password.
# If less than 0 it is the minimum number of lowercase characters in the new
# password.
# lcredit = 0
#
# The maximum credit for having other characters in the new password.
# If less than 0 it is the minimum number of other characters in the new
# password.
# ocredit = 0
#
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
# minclass = 0
#
```

After:

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```
Switch to PowerShell Restart Manage files New session Editor Open in VS Code for the Web
GNU nano 7.2 /etc/security/pwquality.conf *
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
minclass = 3
#
# The maximum number of allowed consecutive same characters in the new password.
# The check is disabled if the value is 0.
# maxrepeat = 0
#
# The maximum number of allowed consecutive characters of the same class in the
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo
```

```
Switch to PowerShell Restart Manage files New session Editor Open in VS Code for the Web
GNU nano 7.2 /etc/security/pwquality.conf *
# difok = 1
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
minlen = 12
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo
```

2. Configure pam_faillock (or pam_tall2, depending on distribution) to lock account after 5 consecutive failed attempts for at least 15 minutes.

```
student@LinuxTargetVM2:~$ pwd
/home/student
student@LinuxTargetVM2:~$ sudo cp /etc/security/faillock.conf /etc/security/faillock.conf.bak
```

Before:

```
# Deny access if the number of consecutive authentication failures
# for this user during the recent interval exceeds n tries.
# The default is 3.
# deny = 3

# The access will be re-enabled after n seconds after the lock out.
# The value 0 has the same meaning as value 'never' - the access
# will not be re-enabled without resetting the faillock
# entries by the 'faillock' command.
# The default is 600 (10 minutes).
# unlock_time = 600
```

After:

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```
GNU nano 7.2 /etc/security/faillock.conf
#
# Deny access if the number of consecutive authentication failures
# for this user during the recent interval exceeds n tries.
# The default is 3.
deny = 5
#
# The length of the interval during which the consecutive
# authentication failures must happen for the user account
# lock out is <replaceable>n</replaceable> seconds.

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo

GNU nano 7.2 /etc/security/faillock.conf
# The value 0 has the same meaning as value `never` - the access
# will not be re-enabled without resetting the faillock
# entries by the `faillock` command.
# The default is 600 (10 minutes).
unlock_time = 900
#
# Root account can become locked as well as regular accounts.
# Enabled if option is present.
even_deny_root

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo
```

Step 3. Test your lockout policy with a series of deliberately failed logins on a test account, and confirm the account locks as expected.

```
cybersec-bootcamp-2026-aug03 [ ~ ]$ ssh student@40.76.125.30
The authenticity of host '40.76.125.30 (40.76.125.30)' can't be established.
ED25519 key fingerprint is SHA256:hAxbjFVHds0Fc0LGLwtLzxsIEc3EZtco4+13qBbaKUI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '40.76.125.30' (ED25519) to the list of known hosts.
student@40.76.125.30's password:
Permission denied, please try again.
student@40.76.125.30's password:
Permission denied, please try again.
student@40.76.125.30's password:
student@40.76.125.30: Permission denied (publickey,password).
cybersec-bootcamp-2026-aug03 [ ~ ]$ ssh student@40.76.125.30
student@40.76.125.30's password:
Permission denied, please try again.
student@40.76.125.30's password:
Permission denied, please try again.
student@40.76.125.30's password:
^Z
[1]+  Stopped                  ssh student@40.76.125.30
cybersec-bootcamp-2026-aug03 [ ~ ]$ ssh student@40.76.125.30
█
```

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```
student@LinuxTargetVM2:~$ sudo faillock --user root
root:
When          Type Source          Valid
2026-08-10 20:26:19 RHOST 41.74.91.200    V
2026-08-10 20:26:24 RHOST 64.235.58.29    V
2026-08-10 20:29:33 RHOST 45.148.10.152   V
2026-08-10 20:29:36 RHOST 45.148.10.152   V
2026-08-10 20:29:40 RHOST 45.148.10.152   V
2026-08-10 20:29:43 RHOST 45.148.10.152   V
2026-08-10 20:29:47 RHOST 45.148.10.152   V
```

After much difficulty we have found that the faillock is in root not student.

Part 3: Audit Logging and Legal Banner

1. Install and enable auditd, and confirm authentication events are being captured in the audit log.

```
student@LinuxTargetVM2:~$ pwd
/home/student
student@LinuxTargetVM2:~$ sudo apt install auditd audispd-plugins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.1build1.1).
The following packages were automatically installed and are no longer required:
  libmbim-utils libqmi-utils libtcl8.6 tcl tcl8.6 usb-modeswitch usb-modeswitch-data
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  audispd-plugins
0 upgraded, 1 newly installed, 0 to remove and 10 not upgraded.
Need to get 38.8 kB of archives.
After this operation, 131 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 audispd-plugins amd64 1:3.1.2-2.1build1.1 [38.8 kB]
Fetched 38.8 kB in 0s (801 kB/s)
Selecting previously unselected package audispd-plugins.
(Reading database ... 99497 files and directories currently installed.)
Preparing to unpack .../audispd-plugins_1%3a3.1.2-2.1build1.1_amd64.deb ...
Unpacking audispd-plugins (1:3.1.2-2.1build1.1) ...
Setting up audispd-plugins (1:3.1.2-2.1build1.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.17.0-1021-azure
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.17.0-1022-azure.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

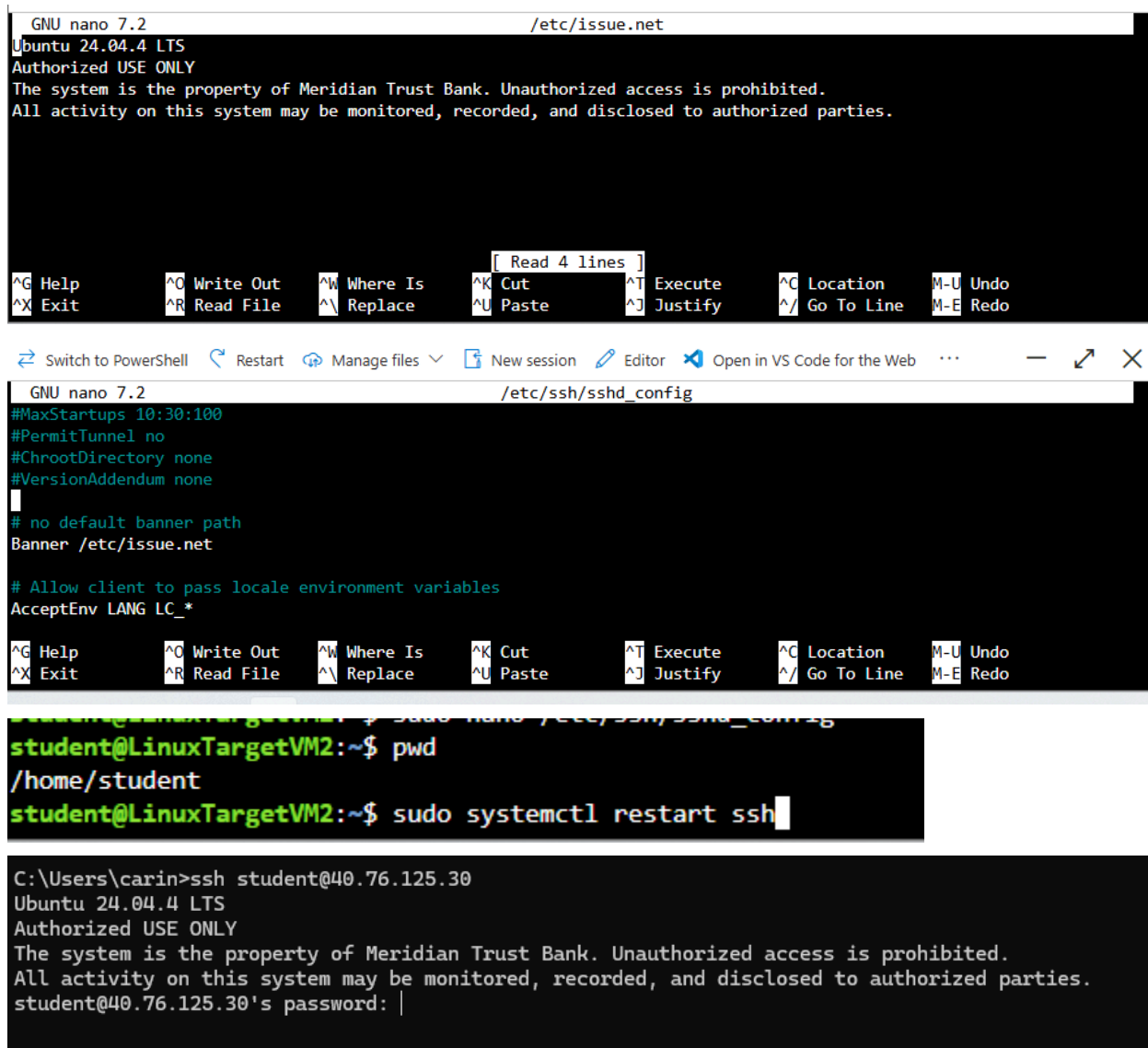
```
student@LinuxTargetVM2:~$ pwd
/home/student
student@LinuxTargetVM2:~$ sudo systemctl enable auditd --now
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
```

```
student@LinuxTargetVM2:~$ pwd
/home/student
student@LinuxTargetVM2:~$ sudo systemctl status auditd
● auditd.service - Security Auditing Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-10 18:32:56 UTC; 1h 15min ago
     Docs: man:auditd(8)
           https://github.com/linux-audit/audit-documentation
  Main PID: 70833 (auditd)
    Tasks: 2 (limit: 9434)
   Memory: 980.0K (peak: 2.2M)
      CPU: 104ms
   CGroup: /system.slice/auditd.service
           └─70833 /sbin/auditd
```

```
time-->Mon Aug 10 19:50:22 2026
type=USER_LOGIN msg=audit(1786391422.822:939): pid=92879 uid=0 auid=1000 ses=906 subj=unconfined msg='op=login id=1000 exe="/usr/sbin/sshd" hostname=71.182.1
40.239 addr=71.182.140.239 terminal=/dev/pts/0 res=success'
```


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2. Configure /etc/issue.net with an authorized-use warning banner, and update sshd_config to display it before login (Banner /etc/issue.net).



```
GNU nano 7.2 /etc/issue.net
Ubuntu 24.04.4 LTS
Authorized USE ONLY
The system is the property of Meridian Trust Bank. Unauthorized access is prohibited.
All activity on this system may be monitored, recorded, and disclosed to authorized parties.

[ Read 4 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo

Switch to PowerShell Restart Manage files New session Editor Open in VS Code for the Web

GNU nano 7.2 /etc/ssh/sshd_config
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none
# no default banner path
Banner /etc/issue.net

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo

student@LinuxTargetVM2:~$ sudo nano /etc/ssh/sshd_config
/home/student
student@LinuxTargetVM2:~$ sudo systemctl restart ssh

C:\Users\carin>ssh student@40.76.125.30
Ubuntu 24.04.4 LTS
Authorized USE ONLY
The system is the property of Meridian Trust Bank. Unauthorized access is prohibited.
All activity on this system may be monitored, recorded, and disclosed to authorized parties.
student@40.76.125.30's password: |
```

Part 4: File Integrity Baseline

1. Install AIDE and initialize its database against the current state of the system.

```
student@LinuxTargetVM2:~$ pwd
/home/student
student@LinuxTargetVM2:~$ sudo apt install aide aide-common -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
aide is already the newest version (0.18.6-2ubuntu0.1).
aide-common is already the newest version (0.18.6-2ubuntu0.1).
aide-common set to manually installed.
The following packages were automatically installed and are no longer required:
  libmbim-utils libqmi-utils libtcl8.6 tcl tcl8.6 usb-modeswitch usb-modeswitch-data
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 10 not upgraded.
student@LinuxTargetVM2:~$
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
aide is already the newest version (0.18.6-2ubuntu0.1).
aide-common is already the newest version (0.18.6-2ubuntu0.1).
aide-common set to manually installed.
The following packages were automatically installed and are no longer required:
  libmbim-utils libqmi-utils libtcl8.6 tcl tcl8.6 usb-modeswitch usb-modeswitch-data
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 10 not upgraded.
student@LinuxTargetVM2:~$ sudo aideinit
Running aide --init...
```

```
student@LinuxTargetVM2:~$ sudo cp /var/lib/aide.db.new /var/lib/aide/aide.db
cp: cannot stat '/var/lib/aide.db.new': No such file or directory
student@LinuxTargetVM2:~$ sudo cp /var/lib/aide/aide.db.new /var/lib/aide/aide.db
student@LinuxTargetVM2:~$
```

2. Make a small, deliberate unauthorized-looking change to a file under /etc

```
student@LinuxTargetVM2:~$ sudo aide --config-check
ERROR: missing configuration (use '--config' '--before' or '--after' command line parameter)
student@LinuxTargetVM2:~$ sudo aide --config=/etc/aide/aide.conf --config-check
student@LinuxTargetVM2:~$ sudo aide --config=/etc/aide/aide.conf --check

^Z
[7]+  Stopped                  sudo aide --config=/etc/aide/aide.conf --check
student@LinuxTargetVM2:~$
```

```
File: /etc/hosts
Size      : 289                               | 358
Mtime     : 2026-08-10 20:00:37 +0000      | 2026-08-10 20:09:58 +0000
Ctime     : 2026-08-10 20:00:37 +0000      | 2026-08-10 20:09:58 +0000
MD5       : FzoVMYkCVeuTePsHIax3lw==      | m19p1yrtqLEf8MOWJFdI5g==
SHA1      : uYZxBWc/3MJ6/s2Az6Z8pfy3KZI=   | Vhy8TskRjNuUfVLGDTquaCLNmQk=
SHA256    : AdsNIZw/LnwrHUwD5V+CnjHREhQafjyG | 10Ie4SVD7ipSgwSV/jibWuJq12LVMIfP
          : +KXOVpunJ4U=                     | eUkRtKfzMRA=
SHA512    : L48YZeX87NomAR+V0857vUYvtcZiY1FJ | 829PQyPjaAR9pf5fhgeUDjFLDiOCI9oU
          : w0pogr8W2bH4nrp03qA6wSGt5OGhv7H    | qLk6hC96/JCDpEdWSoOcJ3aEisoih/WG
          : 5xvgsIP+FSb+vIU52hCCEw==              | R/pJHjxFEGYmDGrKXpNA9g==
RMD160    : s1K5smcmjJRPySa0dcTXL/2JtLA=   | YeetPRXcdLvU/zYQ3S3iDZi7r4w=
TIGER     : cNEaXiViB3mbn+VL3kYJ2k0ur8hkuMIz | x8qtFFL5ajpLTawz85RR0icVnOI2PgV
CRC32     : +WpR6A==                        | IizVEw==
CRC32B    : 1f9CfQ==                       | Zj7j7g==
HAVAL     : RpwJjnniFLly3idXNJft37CJLX7d0ccv | 1Q0AjX84IY82BXkKvBab0r0ngVMH3vow
          : 8oSVJJaSdw+U=                           | iVS0mwGzRBk=
WHIRLPOOL : hZgoJlgAaavAN0Y25KUQBtt5A7q3o1th   | dWmN90rQvJHBT0Y/dcGsN2cb9NrgtUIk
          : Mp205M0NsehgjcrBTvK05sFG8kqtt6g4      | Nd9oRip28/nJ0+cNpDhN2kNb1nNz+4FF
          : E3+xdRwNGH+jHf0a00HrhA==              | UpEIZ7cLcyP6CMU25SE3UA==
GOST      : fR0Vs7nvKTHvxN2j9qpxXTWC68IfQ5e | HI7+1Gr/zuZFzjqzqY0w0/waPiz2GyaZ
          : aDqbqLdN0Ys=                             | YGvKcQ6kJLc=
```

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```
ERROR: missing configuration (use '--config' '--before' or '--after' command line parameter)
student@LinuxTargetVM3:~$ sudo aide -c /etc/aide/aide.conf --check --log-level=info > results@.txt
INFO: initialize rule tree
INFO: define default attribute definitions
INFO: define default groups definitions
INFO: read command line parameters
INFO: (--config): set config file to '/etc/aide/aide.conf'
INFO: (--check): database check command
INFO: (--log-level): set log level to 'info'
INFO: parse configuration
INFO: populate tree
INFO: read old entries from database: file:/var/lib/aide/aide.db
```

Part 5: Build a Compliance-Check Script

```
root@LinuxTargetVM3:~# nano compliance_check.sh
root@LinuxTargetVM3:~# ./compliance_check.sh
-bash: ./compliance_check.sh: Permission denied
root@LinuxTargetVM3:~# chmod +x compliance_check.sh
```

Write a single Bash script (compliance_check.sh) that checks the host against the policy and prints a clear PASS or FAIL for each item. At minimum, it should check:

- That the configured minimum password length meets policy
- That account lockout is configured with the correct threshold
- That auditd is installed and running
- That the SSH login banner is configured
- That AIDE is installed and has an initialized database

```
#!/bin/bash
echo "Meridian Trust Bank - Linux Compliance Check"

MINLEN=$(grep -E "minlen" /etc/security/pwquality.conf | awk -F= '{print $2}' | tr -d ' ')
if [ "$MINLEN" -ge 12 ] 2./dev/null; then
    echo "[PASS] Minimum password length is $MINLEN (policy requires 12+)"
else
    echo "[FAIL] Minimum password length is not set to 12 or higher"
fi

DENY=$(grep -E "deny" /etc/security/faillock.conf | awk -F= '{print $2}' | tr -d ' ')
if [ "$DENY" -le 5 ] 2./dev/null && [ "$DENY" -ge 1 ] 2./dev/null; then
    echo "[PASS] Account lockout threshold is $DENY (policy requires 5 or fewer)"
else
    echo "[FAIL] Account lockout threshold is not configured correctly"
fi

if systemctl is-active --quiet auditd; then
    echo "[PASS] auditd is installed and running"
else
    echo "[FAIL] auditd is not running"
fi

if grep -q "Banner /etc/issue.net" /etc/ssh/sshd_config; then
    echo "[PASS] SSH login banner is configured"
else
    echo "[FAIL] SSH login banner is not configured"
fi

if dpkg -l | grep -q "ii aide" && [ -f /var/lib/aide/aide.db ]; then
    echo "[PASS] AIDE is installed with an initialized database"
else
    echo "[FAIL] AIDE is not installed or has no database"
fi
```

1. Run your finished script and capture its output as evidence

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```
root@LinuxTargetVM3:~# ./compliance_check.sh
Meridian Trust Bank - Linux Compliance Check
[PASS] Minimum password length is 12 (policy requires 12+)
[PASS] Account lockout threshold is 5 (policy requires 5 or fewer)
[PASS] auditd is installed and running
[PASS] SSH login banner is configured
[PASS] AIDE is installed with an initialized database
Check complete
```

2. Fix anything your own script flags as a FAIL before submitting.
 - a. Everything in the script ran ensured nothing has failed

Part 6: Reflection Questions

1. Which policy line was hardest to translate into a technical control, and why?
 - a. The account lockout policy was the hardest control for us to configure and verify. At first, we thought the lockout was not working because we could not see the failed login attempts in the output we were checking. We eventually found that we needed to check the faillock information as root, which showed that the failed attempts were actually being recorded. Once we figured that out, we were able to confirm that the account locked after five failed login attempts as required.
2. Why does a written policy matter even though the technical controls are what actually enforce it? What could go wrong with controls but no documented policy, or a policy with no enforced controls?
 - a. A written policy gives the organization a clear standard for how a system is intended to be secured. Technical controls are in place to enforce said requirements, but without a written policy there would not be a consistent standard for administrators to use as a reference. Having both a written policy and technical controls ensures security expectations are documented and implemented.
3. How does your compliance script change the way this host gets verified in the future compared to a onetime manual check?
 - a. The compliance script makes checking the system much faster and more consistent in the future. Instead of manually opening several configuration files and checking each setting, the script automatically checks the controls and displays a PASS or FAIL result. This also lowers any chance that a compliance check would forget to include one of the requirements.